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System and method for generating a panel infill geometry of a sandwich panel

Abstract

A method of generating a panel infill geometry of a sandwich panel, includes providing a driver mesh representing a panel mid-surface of a sandwich panel. The driver mesh is comprised of a plurality of quadrilateral elements. The method further includes providing a reference unit cell mesh configured to fit exactly within a cube. The reference unit cell mesh is comprised of a unit infill mesh interconnecting a pair of unit face sheet meshes. The method additionally includes mapping a plurality of the reference unit cell meshes respectively onto a plurality of hexahedral elements respectively associated with the plurality of quadrilateral elements, through the use of basis functions defined on each of the plurality of quadrilateral elements in a manner causing adjustment of the size and shape of the plurality of reference unit cell meshes to conform respectively to the plurality of hexahedral elements.

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Background/Summary

FIELD

(1) The present disclosure relates generally to additive manufacturing and, more particularly, to a system and method for generating a panel infill geometry for an additively manufacturable

sandwich panel.

BACKGROUND

(2) A sandwich panel is a structure made up of a pair of face sheets, interconnected by an inner core. The inner core is typically a relatively lightweight material or a lightweight structural arrangement, and results in a lightweight panel having a relatively high specific bending stiffness. Sandwich panels can be fabricated using additive manufacturing, which involves the successive application and solidification of layers of material on top of each other.

(3) The conventional process of additively manufacturing a sandwich panel involves fabricating the inner core based on a prior art infill geometry. A prior art infill geometry has an orthogonally-repeating structural pattern of uniform size and shape. During additive manufacturing, only the portion of the structural pattern that lies between the boundaries of the face sheets is fabricated. This may result in irregular and inconsistent topology at the interfaces between the inner core and the face sheets. The shape at certain interface locations can present manufacturing challenges, and/or can result in stress concentrations when the sandwich panel is loaded.

(4) As can be seen, there exists a need in the art for a method of generating a panel infill geometry for an inner core of a sandwich panel in a manner that results in a consistent footprint pattern at the interface locations between the inner core and the face sheets.

SUMMARY

(5) The above-noted needs associated with manufacturing sandwich panels are addressed by the present disclosure, which provides a method of generating a panel infill geometry for interconnecting a pair of face sheets of a sandwich panel. The method includes providing a driver mesh representing a panel mid-surface of a sandwich panel. The driver mesh is comprised of a plurality of quadrilateral elements. The method also includes providing a reference unit cell mesh configured to fit exactly within a cube. The reference unit cell mesh is comprised of a unit infill mesh interconnecting a pair of unit face sheet meshes. Additionally, the method includes mapping a plurality of the reference unit cell meshes respectively onto a plurality of hexahedral elements respectively associated with the plurality of quadrilateral elements, through the use of basis functions defined on each of the plurality of quadrilateral elements in a manner causing adjustment of the size and shape of the plurality of reference unit cell meshes to conform respectively to the plurality of hexahedral elements, and resulting in a plurality of mapped unit cell meshes collectively forming a panel infill mesh having a panel infill geometry interconnecting a pair of face sheet meshes.

(6) Also disclosed is a processor-based system for generating a panel infill geometry for interconnecting a pair of face sheets of a sandwich panel. The processor-based system comprises a memory device configured to store a driver mesh representing a panel mid-surface of a sandwich panel. The driver mesh is comprised of a plurality of quadrilateral elements. The memory device is also configured to store a reference unit cell mesh configured to fit exactly within a cube. The processor-based system further comprises a unit cell mapping module configured to map a plurality of the reference unit cell meshes respectively onto a plurality of hexahedral elements respectively associated with the plurality of quadrilateral elements, through the use of basis functions defined on each of the plurality of quadrilateral elements in a manner causing adjustment of the size and shape of the plurality of reference unit cell meshes to conform respectively to the plurality of hexahedral elements, and resulting in a plurality of mapped unit cell meshes collectively forming a panel infill mesh having a panel infill geometry interconnecting a pair of face sheet meshes.

(7) In addition, disclosed is a sandwich panel, comprising a first face sheet, a second face sheet, and a panel infill structure comprising an array of unit infill structures arranged in one or more layers connecting the first face sheet to the second face sheet, and each unit infill structure has a main axis that is locally normal to an imaginary mid-surface midway between the first face sheet and the second face sheet.

(8) The features, functions, and advantages that have been discussed can be achieved independently

in various versions of the disclosure or may be combined in yet other versions, further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure can be better understood with reference to the following detailed description, taken in conjunction with the accompanying drawings, which illustrate preferred and exemplary versions, but which are not necessarily drawn to scale. The drawings are examples, and are not meant as limitations on the description or the claims.
- (2) FIG. 1 is a perspective view of an example of sandwich panel, comprised of first and second face sheets, and an inner core;
- (3) FIG. 2 is a perspective view of an example of the orthogonal pattern of geometry from which the prior art infill geometry is generated;
- (4) FIG. 3 is a perspective view of the prior art sandwich panel of FIG. 1 showing the prior art infill geometry contained between the first and second face sheets;
- (5) FIG. 4 is a perspective view of a further example of a prior art sandwich panel having an inner core of prior art infill geometry;
- (6) FIG. 5 is a magnified view of the portion of the sandwich panel identified by reference numeral 5 of FIG. 4, illustrating the irregular and inconsistent topology at the interfaces between the inner core and the face sheets;
- (7) FIG. 6 is a perspective view of an example of a presently-disclosed sandwich panel comprised of a panel infill structure between first and second face sheets, and showing a plurality of unit cell structures that make up the panel infill structure;
- (8) FIG. 7 is a magnified view of the portion of the sandwich panel identified by reference numeral 7 of FIG. 6, illustrating the consistent topology at the interfaces between the panel infill structure and the first and second face sheets;
- (9) FIG. 8 is a flowchart of operations included in the method of generating a panel infill geometry of a panel infill structure of a sandwich panel;
- (10) FIG. 9 is a perspective view of an example of a digital model (e.g., a 3D computer-aided-design (CAD) model) of a panel mid-surface between first and second face sheets of a sandwich panel;
- (11) FIG. 10 is a perspective view of the panel mid-surface of FIG. 9 after the panel mid-surface has been meshed into a driver mesh comprised of a plurality of quadrilaterals elements;
- (12) FIG. 11 is a perspective view of an example of a cube;
- (13) FIG. 12 is a perspective view of an example of a reference unit cell mesh configured to fit exactly within the cube of FIG. 11, and comprised of trimmed parametric surfaces that have been meshed;
- (14) FIG. 13 shows a further example of a reference unit cell mesh configured to fit exactly within the cube of FIG. 11, and in which the reference unit cell mesh is comprised of trimmed parametric surfaces enclosing a volume, which has been meshed with solid elements (e.g., tetrahedral elements);
- (15) FIG. 14 is a flowchart of operations included in a method of mapping a reference unit cell mesh onto a hexahedral element associated with a linear quadrilateral element of a driver mesh;
- (16) FIG. 15 is a magnified view of the portion of the driver mesh of FIG. 10, comprised of linear (i.e., 4-noded) quadrilateral elements, and illustrating the 4 corner nodes of one of the linear quadrilateral elements;
- (17) FIG. 16 shows an example of a $2 \times 2 \times 2$ cube in isoparametric space;
- (18) FIG. 17 shows an example of the cube of FIG. 16 mapped onto the hexahedral element

associated with the linear quadrilateral element of FIG. 15;

(19) FIG. 18A shows a plurality of level set contours of a basis function defined on a first corner node of a 4-noded linear quadrilateral element;

(20) FIG. 18B shows a plurality of level set contours of a basis function defined on a second corner node of the 4-noded linear quadrilateral element;

(21) FIG. 18C shows a plurality of level set contours of a basis function defined on a third corner node of the 4-noded linear quadrilateral element;

(22) FIG. 18D shows a plurality of level set contours of a basis function defined on a fourth corner node of the 4-noded linear quadrilateral element;

(23) FIG. 19 is an example of a reference unit cell mesh fitting exactly within the cube of FIG. 16;

(24) FIG. 20 shows the reference unit cell mesh of FIG. 19 mapped onto the hexahedral element of FIG. 17;

(25) FIG. 21 is a flowchart of operations included in a method of mapping a reference unit cell mesh onto a hexahedral element associated with an 8-noded quadratic quadrilateral element;

(26) FIG. 22 is an example of the driver mesh of FIG. 10 comprised of quadratic quadrilateral elements, and illustrating the 4 corner nodes and 4 mid-side nodes of one of the quadratic quadrilateral elements;

(27) FIG. 23 is an example of a cube in isoparametric space;

(28) FIG. 24 shows an example of the cube of FIG. 23 mapped onto a hexahedral element associated with the quadratic quadrilateral element of FIG. 22;

(29) FIG. 25A shows a plurality of level set contours of a basis function defined on a first corner node of an 8-noded quadratic quadrilateral element;

(30) FIG. 25B shows a plurality of level set contours of a basis function defined on a first mid-side node of the 8-noded quadratic quadrilateral element;

(31) FIG. 25C shows a plurality of level set contours of a basis function defined on a second corner node of the 8-noded quadratic quadrilateral element;

(32) FIG. 25D shows a plurality of level set contours of a basis function defined on a second mid-side node of the 8-noded quadratic quadrilateral element;

(33) FIG. 25E shows a plurality of level set contours of a basis function defined on a third corner node of the 8-noded quadratic quadrilateral element;

(34) FIG. 25F shows a plurality of level set contours of a basis function defined on a third mid-side of the 8-noded quadratic quadrilateral element;

(35) FIG. 25G shows a plurality of level set contours of a basis function defined on a fourth corner node of the 8-noded quadratic quadrilateral element;

(36) FIG. 25H shows a plurality of level set contours of a basis function defined on a fourth mid-side node of the 8-noded quadratic quadrilateral element;

(37) FIG. 26 is an example of a reference unit cell mesh fitting exactly within the cube of FIG. 23;

(38) FIG. 27 shows the reference unit cell mesh of FIG. 26 mapped onto the hexahedral element of FIG. 24;

(39) FIG. 28 is a perspective view of a sandwich panel having a panel infill geometry comprised of a plurality of unit cell meshes, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. 10;

(40) FIG. 29 is an alternative perspective view of the sandwich panel of FIG. 28, and illustrating the consistent topology at the interfaces between the panel infill geometry and the face sheets;

(41) FIG. 30 is a side view of an example of an additive manufacturing apparatus for fabricating a sandwich panel having a panel infill geometry generated using the presently-disclosed system and method;

(42) FIG. 31 is a section view of the additive manufacturing apparatus of FIG. 30;

(43) FIG. 32 is a perspective view of an example of a driver mesh;

(44) FIG. 33 is a perspective view of an example of a unit cell geometry configured as a

honeycomb;

(45) FIG. **34** is a perspective view of a reference unit cell mesh comprised of shell elements, and representing the reference unit cell geometry of FIG. **33**;

(46) FIG. **35** is a perspective view of a sandwich panel mesh (shown without the face sheet meshes) comprised of shell elements, and generated using the reference unit cell mesh of FIG. **34**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(47) FIG. **36** is a perspective view of a sandwich panel mesh comprised of solid elements, and generated using a solid meshed version (not shown) of the reference unit cell geometry of FIG. **33**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(48) FIG. **37** is a perspective view of an example of a reference unit cell mesh configured as a gyroid;

(49) FIG. **38** is a perspective view of a reference unit cell mesh comprised of shell elements, and representing the reference unit cell geometry of FIG. **37**;

(50) FIG. **39** is a perspective view of a sandwich panel mesh (shown without the face sheet meshes) comprised of shell elements, and generated using the reference unit cell mesh of FIG. **38**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(51) FIG. **40** is a perspective view of a sandwich panel mesh comprised of solid elements, and generated using a solid meshed version of the reference unit cell geometry (see FIG. **13**) of FIG. **37**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(52) FIG. **41** is a perspective view of an example of a reference unit cell mesh in a Schwarz-P configuration;

(53) FIG. **42** is a perspective view of a reference unit cell mesh comprised of shell elements, and representing the reference unit cell geometry of FIG. **41**;

(54) FIG. **43** is a perspective view of a sandwich panel mesh (shown without the face sheet meshes) comprised of shell elements, and generated using the reference unit cell mesh of FIG. **42**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(55) FIG. **44** is a perspective view of a sandwich panel mesh comprised of solid elements, and generated using a solid meshed version (not shown) of the reference unit cell geometry of FIG. **41**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(56) FIG. **45** is a perspective view of an example of a reference unit cell mesh in a waffle configuration;

(57) FIG. **46** is a perspective view of a reference unit cell mesh comprised of shell elements, and representing the reference unit cell geometry of FIG. **45**;

(58) FIG. **47** is a perspective view of a sandwich panel mesh (shown without the face sheet meshes) comprised of shell elements, and generated using the reference unit cell mesh of FIG. **46**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(59) FIG. **48** is a perspective view of a sandwich panel mesh comprised of solid elements, and generated using a solid meshed version (not shown) of the reference unit cell geometry of FIG. **45**, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. **32**;

(60) FIG. **49** is a perspective view of an example of a reference unit cell mesh in a body-centered-cubic lattice configuration;

(61) FIG. **50** is a perspective view of a reference unit cell mesh comprised of shell elements, and

representing the reference unit cell geometry of FIG. 49;

(62) FIG. 51 is a perspective view of a sandwich panel mesh (shown without the face sheet meshes) comprised of shell elements, and generated using the reference unit cell mesh of FIG. 50, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. 32;

(63) FIG. 52 is a perspective view of a sandwich panel mesh comprised of solid elements, and generated using a solid meshed version (not shown) of the reference unit cell geometry of FIG. 49, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. 32;

(64) FIG. 53 is a perspective view of an example of a multi-layer unit cell geometry;

(65) FIG. 54 is a perspective view of a reference unit cell mesh representing the multi-layer unit cell geometry of FIG. 53;

(66) FIG. 55 is a perspective view of a sandwich panel mesh (shown without the face sheet meshes) comprised of shell elements, and generated using the reference unit cell mesh of FIG. 54, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. 32;

(67) FIG. 56 is a perspective view of a sandwich panel mesh comprised of solid elements, and generated using a solid meshed version of the reference unit cell geometry of FIG. 53, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. 32;

(68) FIG. 57 is a perspective view of an example of an unstructured driver mesh;

(69) FIG. 58 is a perspective view of an example of a unit cell geometry in a Schwarz-P configuration;

(70) FIG. 59 is a perspective view of an example of a sandwich panel mesh (shown without the panel face sheet meshes) comprised of shell elements, and generated using the reference unit cell mesh of FIG. 58, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the unstructured driver mesh of FIG. 57;

(71) FIG. 60 is a perspective view of an example of a driver mesh that is spatially graded;

(72) FIG. 61 is a perspective view of an example of a unit cell geometry configured as a waffle with center holes;

(73) FIG. 62 is a perspective view of an example of a sandwich panel mesh comprised of solid elements, and generated using a solid meshed version (not shown) of the unit cell geometry of FIG. 61, mapped onto the hexahedral elements respectively associated with the quadrilateral elements of the spatially-graded driver mesh of FIG. 60;

(74) FIG. 63 is a perspective view of an example of a constant-thickness sandwich panel having a panel infill geometry based on a Schwarz-P configuration of the unit cell geometry;

(75) FIG. 64 is a perspective view of an example of a variable-thickness sandwich panel having a panel infill geometry based on the Schwarz-P configuration of the unit cell geometry;

(76) FIG. 65 is a perspective view of an example of a constant-thickness sandwich panel having a panel infill geometry based on a body-centered-cubic lattice configuration of the unit cell geometry;

(77) FIG. 66 is a perspective view of an example of a variable-thickness sandwich panel having a panel infill geometry based on the body-centered-cubic lattice configuration of the unit cell geometry;

(78) FIG. 67 is a side view of an example of a nose section of an aircraft fuselage;

(79) FIG. 68 is a perspective view of a driver mesh representing the panel mid-surface of a portion of the aircraft fuselage of FIG. 67;

(80) FIG. 69 is a perspective view of an example of a replacement sandwich panel for the aircraft fuselage, generated by mapping a unit cell geometry onto the hexahedral elements respectively associated with the quadrilateral elements of the driver mesh of FIG. 68;

(81) FIG. 70 is a block diagram of an example of a processor-based system for implementing the presently-disclosed method for generating a panel infill geometry of a sandwich panel; and (82) FIG. 71 shows an example of a user interface for the processor-based system of FIG. 70. (83) The figures shown in this disclosure represent various aspects of the versions presented, and only differences will be discussed in detail.

DETAILED DESCRIPTION

(84) Disclosed versions will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all of the disclosed examples are shown. Indeed, several different examples or versions may be provided and should not be construed as limited to the examples set forth herein. Rather, these examples are provided so that this disclosure will be thorough and fully convey the scope of the disclosure to those skilled in the art.

(85) This specification includes references to “some examples,” “one example,” or “an example.” Instances of the phrases “some examples,” “one example” or “an example” do not necessarily refer to the same example. Particular features, structures, or characteristics may be combined in any suitable manner consistent with this disclosure.

(86) As used herein, “comprising” is an open-ended term, and as used in the claims, this term does not foreclose additional structures or steps.

(87) As used herein, “configured to” means various parts or components may be described or claimed as “configured to” perform a task or tasks. In such contexts, “configured to” is used to connote structure by indicating that the parts or components include structure that performs those task or tasks during operation. As such, the parts or components can be said to be configured to perform the task even when the specified part or component is not currently included.

(88) As used herein, an element or step recited in the singular and preceded by the word “a” or “an” should be understood as not necessarily excluding the plural of the elements or steps.

(89) As used herein, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items may be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item may be a particular object, a thing, or a category.

(90) Referring now to the drawings, shown in FIG. 1 is an example of a prior art sandwich panel **100**, which has a first face sheet **102**, a second face sheet **104**, and an inner core **106** (indicated as a cross-hatched region), as shown in FIG. 3. The inner core **106** has a prior art infill geometry **166**, which has an orthogonally repeating structural pattern of uniform size and shape. In preparation for additively manufacturing the prior art sandwich panel **100**, conventional software computes the portion of the structural pattern that lies between the boundaries of the first and second face sheets **102**, **104**. As shown in FIG. 3, this results in irregular shapes with inconsistent topology at the infill/face sheet interfaces **110** between the inner core **106** and the first and second face sheets **102**, **104** of the additively manufactured prior art sandwich panel **100**. As mentioned above, the shape of certain infill/face sheet interface **110** locations can result in localized stress concentrations. FIGS. 4-5 show another example of a prior art sandwich panel **100**, and the irregular and inconsistent topology at the infill/face sheet interfaces **110** between the inner core **106** and the first and second face sheets **102**, **104**.

(91) Referring now to FIGS. 6-7, shown is an example of a sandwich panel **120** fabricated using the presently disclosed system and method. The sandwich panel **120** of FIGS. 6-7 is comprised of a first face sheet **102**, a second face sheet **104**, and a panel infill structure **164** comprised of a plurality of unit cell structures **206**. As described in greater detail below, the panel infill structure **164** has a panel infill geometry **166** in which each unit cell structure **206** has a main axis **212** that is locally normal to an imaginary mid-surface (not shown) midway between the first and second face sheets **102**, **104**. Advantageously, such an arrangement results in a consistent topology at the infill/face sheet interfaces **110** between the panel infill structure **164** and the first and second face

sheets **102**, **104**, as shown in FIG. 7.

(92) Referring to FIG. 8, with additional reference to FIGS. 9-71, shown in FIG. 8 is a flow chart of operations included in a method **700** of generating a panel infill geometry **166** for a sandwich panel **120**, similar to the examples shown in FIGS. 28, 29, 36, 40, 44, 48, 52, 56, 59, 62-66, and 69. Step **702** of the method **700** comprises providing a driver mesh **142** (FIG. 10), which represents a panel mid-surface **126** (FIG. 9) midway between the first and second face sheets **102**, **104** (FIGS. 28-29) of a sandwich panel **120** (FIGS. 28-29). The driver mesh **142** is made up of a plurality of quadrilateral elements **148**. Each quadrilateral element **148** has driver mesh nodes **156** and edges **162**. A driver mesh **142** can be constructed by meshing a driver digital model **128** (e.g., a computer-aided-design model or CAD model) of the panel mid-surface **126** of a sandwich panel **120** (FIGS. 28-29). FIG. 9 shows an example of a driver digital model **128** of a panel mid-surface **126** of a sandwich panel **120** to be additively manufactured. FIG. 10 shows an example of a driver mesh **142** generated by meshing the driver digital model **128** of FIG. 9.

(93) In some examples, step **702** comprises providing the driver mesh **142** as a structured driver mesh **144**. A structured driver mesh **144** may be described as having a pattern of quadrilateral elements **148** which can be mapped into a square domain with uniform rows and columns (i.e., utilizing a transfinite interpolation method of mesh generation), as exemplified in FIG. 10. In other examples, step **702** comprises providing the driver mesh **142** as an unstructured driver mesh **146** (FIG. 57). In an unstructured driver mesh **146**, the quadrilateral elements **148** are a general arrangement of quadrilateral elements **148** connected together at common driver mesh nodes **156**, where there is no possible mapping into a square domain with uniform rows and columns. FIG. 57, described in greater detail below, shows an example of an unstructured driver mesh **146**. In still other examples, step **702** comprises providing a driver mesh **142** (i.e., either structured or unstructured) that is spatially graded over at least one region of the panel mid-surface **126**, as shown in the example of FIG. 60, and described in greater detail below.

(94) A driver mesh **142** can be made up of linear quadrilateral elements **150** (e.g., FIG. 15). Each linear quadrilateral element **150** has 4 corner nodes **158**, such that each edge **162** of each linear quadrilateral element **150** consists of 2 corner nodes **158**, as shown in FIG. 15 and described in greater detail below. Alternatively, a driver mesh **142** can be made up of quadratic quadrilateral elements **152** (e.g., FIG. 22), which allows for curvature within each quadrilateral element **148**. Each quadratic quadrilateral element **152** has 4 corner nodes **158** and 4 mid-side nodes **160**, such that each edge **162** of each quadratic quadrilateral element **152** consists of 2 corner nodes **158** and 1 mid-side node **160**, as shown in FIG. 22 and described in greater detail below.

(95) Referring to FIG. 11-13, step **704** of the method **700** comprises providing a reference unit cell mesh **300** (FIG. 12). The reference unit cell mesh **300** is configured to fit exactly within a $2 \times 2 \times 2$ cube **202** (FIG. 11). The cube **202** has 6 square faces **204**. The reference unit cell mesh **300** is described in isoparametric space **200** according to an orthonormal coordinate system having η , ξ , and ζ axes, and origin (0,0,0) at the center of the cube **202**, as shown in FIG. 11. In addition, the reference unit cell mesh **300** has a main axis **212** that is coincident with the ζ axis. The reference unit cell mesh is comprised of a unit infill mesh **314** interconnecting a pair of unit face sheet meshes **318** on opposite ends of the unit infill mesh **314**. Each of the unit face sheet meshes **318** is the same size and shape as the square face respectively defining the top and bottom of the cube.

(96) In the example of FIG. 12, the reference unit cell mesh **300** is given as a surface representation comprised of shell elements **306**. In the example of FIG. 13, the reference unit cell mesh **300** is given as a solid representation comprised of solid elements **308**. In both FIGS. 12 and 13, these reference unit cell meshes **300** illustrate the example of a gyroid **216** configuration. Alternatively, in other examples, step **704** comprises providing the reference unit cell mesh **300** with a unit cell geometry **210** having one of the following configurations: Schwarz-P **214** (FIG. 41), body-centered-cubic lattice **218** (FIG. 49), honeycomb **220** (FIG. 33), or waffle **222** (FIG. 45), as shown in FIGS. 33-52 and described below.

(97) Referring briefly to FIGS. 53-54, a reference unit cell mesh 300 can be provided in a multi-layer unit cell geometry 226, represented by a multi-layer unit cell mesh 228. For example, FIG. 53 shows a multi-layer unit cell geometry 226 comprised of a multi-layer unit infill geometry 230 interconnecting a pair of unit face sheet geometries 316 respectively on the top and bottom of the multi-layer unit infill geometry 230. In the example shown, the multi-layer unit infill geometry 230 is comprised of 3 layers 234 of unit infill geometries 310 stacked on top of each other.

(98) FIG. 54 shows a reference unit cell mesh 300 representing the multi-layer unit cell geometry 226 of FIG. 53. The reference unit cell mesh 300 is a multi-layer unit cell mesh 228 comprised of a multi-layer unit infill mesh 232 interconnecting a pair of unit face sheet meshes 318 respectively on the top and bottom of the multi-layer unit infill mesh 232. The multi-layer unit infill mesh 232 is comprised of 3 layers 234 of unit infill meshes 314. Each unit infill mesh 314 has a gyroid 216 configuration, which is triply periodic. Each layer 234 contains 3 rows of the unit infill meshes 314 in the η direction, and each row contains 3 unit infill meshes 314 in the ξ direction. The main axes 212 (FIG. 38) of the unit infill geometries 310 are aligned with each other. Each unit infill geometry 310 is periodic in the ζ direction of the cube (e.g., FIG. 11), allowing the unit infill geometries 310 to be interconnected.

(99) A multi-layer unit cell geometry 226 can be provided in alternative configurations, and is not limited to the configuration shown in FIGS. 53-54. In this regard, a multi-layer unit cell geometry 226 can include 2 or more layers 234 of unit infill geometries 310 stacked on top of each other. Each layer 234 can include 1 or more rows of unit infill meshes 314 in the η direction, and each row can include a corresponding number of unit infill meshes 314 in the ξ direction.

(100) A reference unit cell mesh 300 can be provided with other configurations not shown. In this regard, a reference unit cell mesh 300 can be custom-designed by first using a CAD software program to generate a unit infill geometry 310 comprised of trimmed parametric surfaces. After generating the unit infill geometry 310, the trimmed parametric surfaces can be meshed to generate a reference unit cell mesh 300.

(101) A reference unit cell mesh 300 is made up of cell mesh elements having a plurality of cell mesh nodes 304. For example, the reference unit cell mesh 300 is comprised of either shell elements 306 (FIG. 12) or solid elements 308 (e.g., tetrahedral elements—e.g., FIG. 13). For a reference unit cell mesh 300 comprised of shell elements 306 as shown in FIG. 12, step 704 of providing the reference unit cell mesh 300 comprises: constructing the reference unit cell mesh 300 as one or more trimmed parametric surfaces, followed by meshing the one or more trimmed parametric surfaces, to thereby generate the reference unit cell mesh 300 comprised of shell elements 306. For a reference unit cell mesh 300 comprised of solid elements 308 as shown in FIG. 13, step 704 of providing the reference unit cell mesh 300 comprises: constructing the reference unit cell mesh 300 as a solid structure 208 (e.g., FIG. 37) represented by one or more trimmed parametric surfaces enclosing a volume, followed by meshing the volume, to thereby generate the reference unit cell mesh 300 comprised of solid elements 308.

(102) In some examples, step 704 of providing the reference unit cell mesh 300 comprises providing or constructing the reference unit cell mesh 300 to be doubly periodic (i.e., cell mesh nodes 304 belonging to the reference unit cell meshes 300 are in the set custom character.sup.2) or, optionally, triply periodic (i.e., cell mesh nodes 304 belonging to the reference unit cell meshes 300 are in the set custom character.sup.3). In other words, for the doubly periodic case, the reference unit cell mesh 300 is periodic in the η and ξ directions, with the pattern of cell mesh nodes 304 on the square face 204 at $\Theta=1$ being the same as the pattern of cell mesh nodes 304 on the square face 204 at $\Theta=-1$, and the pattern of cell mesh nodes 304 on the square face 204 at $\xi=1$ being the same as the pattern of cell mesh nodes 304 on the square face 204 at $\xi=-1$. For the triply periodic case, the reference unit cell mesh 300 is periodic in the η , ξ and ζ directions, with the pattern of cell mesh nodes 304 on the square face 204 at $\eta=1$ being the same as the pattern of cell mesh nodes 304 on the square face 204 at $\eta=-1$, and the pattern of cell mesh nodes 304 on the

square face **204** at $\xi=1$ being the same as the pattern of cell mesh nodes **304** on the square face **204** at $\xi=1$, and the pattern of cell mesh nodes **304** on the square face **204** at $\xi=1$ being the same as the pattern of cell mesh nodes **304** on the square face **204** at $\xi=-1$. For example, the gyroid **216** configuration of the reference unit cell mesh **300** in FIGS. **12-13** is triply periodic, which means that the reference unit cell mesh **300** is periodic with respect to the η , ξ and ζ directions (i.e., the principal directions) of the cube **202** of FIG. **11**.

(103) In some examples (e.g., FIGS. **41** and **49**) described below, step **704** comprises constructing the reference unit cell mesh **300** with cell branches **224**, each of which terminates at a square face **204** (FIG. **11**) of the cube **202** (FIG. **11**).

(104) When reference unit cell meshes **300** are mapped onto a plurality of hexahedral elements **154** (e.g., FIG. **17**) respectively associated with the plurality of quadrilateral elements **148** of a driver mesh **142** as described below, the periodic nature of the reference unit cell mesh **300** results in mapped unit cell meshes **302** (e.g., FIGS. **20** and **27**) which can be stitched together to form a continuous stitched mesh **130** (i.e., a sandwich panel mesh **132**—e.g., FIG. **28-29**).

(105) Step **706** of the method **700** comprises mapping a plurality of the reference unit cell meshes **300** respectively onto a plurality of hexahedral elements **154** (e.g., FIG. **17**) respectively associated with the plurality of quadrilateral elements **148** of the driver mesh **142** (e.g., FIG. **10**), through the use of basis functions defined on each of the plurality of quadrilateral elements **148**. As described in greater detail below, the process of mapping via basis functions is performed in a manner causing adjustment of the size and shape of the plurality of reference unit cell meshes **300** (e.g., FIG. **19**) to conform respectively to the size and shape of the plurality of hexahedral elements **154** (e.g., FIG. **20**), and resulting in a plurality of mapped unit cell meshes **302** that fit respectively within the plurality of hexahedral elements **154** associated with each of the plurality of quadrilateral elements **148** of the driver mesh **142**. As mentioned above, the reference unit cell mesh **300** is described in isoparametric space **200** according to an orthonormal coordinate system (e.g., having η , ξ , ζ axes) as shown in the example of FIG. **16**. The driver mesh **142** is described in real space **140** according to a real space coordinate system (e.g., having x , y , z axes) as shown in the example of FIG. **17**. After mapping is complete, the result is a panel infill geometry **166** comprised of a plurality of mapped unit infill meshes **314** interconnecting the first face sheet **102** to the second face sheet **104** (e.g., FIGS. **28-29**). Advantageously, the presently-disclosed mapping process results in consistent footprint (i.e., having the same topology) at the infill/face sheet interfaces **110** (FIGS. **28-29**) between the panel infill geometry **166** and the first and second face sheets **102**, **104**.

(106) Referring particularly now to FIGS. **14-20**, step **706** of mapping a reference unit cell mesh **300** is described below with regard to the method **800** of FIG. **14**, which is directed toward mapping a reference unit cell mesh **300** onto the hexahedral elements **154** associated with the linear quadrilateral elements **150** of a driver mesh **142**. FIG. **15** shows a portion of a driver mesh **142** comprised of linear quadrilateral elements **150**. As mentioned above, each linear quadrilateral element **150** has 4 corner nodes **158**, which are identified as p.sub.1 . . . p.sub.4, in real space **140**. FIG. **16** shows a cube **202** in isoparametric space **200**, and the corresponding location of the nodes p.sub.1 . . . p.sub.4 on the cube **202**. FIG. **17** shows a hexahedral element **154** associated with the linear quadrilateral element **150** of FIG. **15**. The height or thickness of the hexahedral element **154** (e.g., at each corner node p.sub.1 . . . p.sub.4) is the local panel thickness t for the sandwich panel **120**, which is specified prior to initiating the mapping process. In one example, the sandwich panel **120** is a constant-thickness sandwich panel **122** (e.g., FIGS. **63** and **65**). Alternatively, the sandwich panel **120** is a variable-thickness sandwich panel **124** (e.g., FIGS. **64** and **66**), in which case the local panel thickness t is different in at least two locations on at least one of the quadrilateral elements **148** contained within the driver mesh **142**, as described below.

(107) The mapping of a reference unit cell mesh **300** onto a plurality of hexahedral elements **154** respectively associated with the plurality of linear quadrilateral elements **150** (FIG. **15**) of a driver

mesh **142** (FIG. **15**) is performed through the use of linear basis functions defined on the linear quadrilateral element **150**. Linear basis functions describe the value of a point of interest within a region, using a weighted combination of values at points around the point of interest. FIG. **18** shows 4 groups of level set contours respectively belonging to the 4 linear basis functions $N_{sub.1} \dots N_{sub.4}$ defined on a 4-noded linear quadrilateral element **150**. The level set contours indicate the weight of a given corner node **158** across the 4-noded linear quadrilateral element **150**. FIG. **19** shows an example of a reference unit cell mesh **300** having a gyroid **216** configuration. FIG. **20** shows the reference unit cell mesh **300** of FIG. **19** after being mapped onto the hexahedral element **154** (e.g., FIG. **17**) associated with the linear quadrilateral element **150** of the driver mesh **142**. (108) In FIG. **14**, the method **800** of mapping the reference unit cell mesh **300** onto each of the plurality of hexahedral elements respectively associated with the plurality of linear quadrilateral elements **150** in a driver mesh **142** comprises sequentially performing the below-described steps **802** and **804** for each linear quadrilateral element **150** of the driver mesh **142**. Step **802** comprises computing the nodal averaged normal vectors $n_{sub.1} \dots n_{sub.4}$ (FIG. **17**) of the corner nodes **158** $p_{sub.1} \dots p_{sub.4}$ of the linear quadrilateral element **150** in real space **140**, using the following equations:

$$n_{sub.x} = \sum_{i=1}^{sup.k} A_{sub.in.sub.i.sub.x} \quad (\text{Equation 602})$$

$$n_{sub.y} = \sum_{i=1}^{sup.k} A_{sub.in.sub.i.sub.y} \quad (\text{Equation 604})$$

$$n_{sub.z} = \sum_{i=1}^{sup.k} A_{sub.in.sub.i.sub.z} \quad (\text{Equation 606})$$

wherein: k is the number of elements (i.e., linear quadrilateral elements **150**) connected to a given corner node **158**; $A_{sub.i}$ is either the area of the $i_{sup.th}$ connecting element (for area weighted averaged normals) or 1 (for uniformly weighted averaged normals); $n_{sub.i.sub.x}$, $n_{sub.i.sub.y}$ and $n_{sub.i.sub.z}$ are the x , y and z components of the $i_{sup.th}$ connecting element normal respectively; and $n_{sub.x}$, $n_{sub.y}$ and $n_{sub.z}$ are the x , y and z components of the nodal averaged normal vector respectively.

(109) Step **804** comprises sequentially performing the following computations for each cell mesh node **304** of the reference unit cell mesh **300**. Initially, step **804** includes computing the linear basis functions $N_{sub.1} \dots N_{sub.4}$ of the cell mesh node **304**, using the following equations:

$$(110) \quad N_1 = \frac{1}{4}(1 - \eta)(1 - \xi) \quad (\text{Equation 608}) \quad N_2 = \frac{1}{4}(1 + \eta)(1 - \xi) \quad (\text{Equation 610})$$

$$N_3 = \frac{1}{4}(1 + \eta)(1 + \xi) \quad (\text{Equation 612}) \quad N_4 = \frac{1}{4}(1 - \eta)(1 + \xi) \quad (\text{Equation 614})$$

wherein η and ξ (e.g., see FIG. **16**) are the coordinates of the cell mesh node **304** in isoparametric space **200**.

(111) After computing the linear basis functions $N_{sub.1} \dots N_{sub.4}$, step **804** comprises computing the normal vector $n = (n_{sub.x}, n_{sub.y}, n_{sub.z})_{sup.T}$ of the panel mid-surface **126** at the position of the cell mesh node **304** in real space **140**, using the following equations:

$$n_{sub.x} = \sum_{i=1}^{sup.4} N_{sub.in.sub.i.sub.x} \quad (\text{Equation 616})$$

$$n_{sub.y} = \sum_{i=1}^{sup.4} N_{sub.in.sub.i.sub.y} \quad (\text{Equation 618})$$

$$n_{sub.z} = \sum_{i=1}^{sup.4} N_{sub.in.sub.i.sub.z} \quad (\text{Equation 620})$$

wherein $n_{sub.i.sub.x}$, $n_{sub.i.sub.y}$, and $n_{sub.i.sub.z}$ are respectively the x , y , and z components (in the real coordinate system—e.g., FIG. **17**) of the nodal averaged normal vectors $n_{sub.1} \dots n_{sub.4}$ at the nodes $p_{sub.1} \dots p_{sub.4}$ respectively.

(112) After computing $n_{sub.i.sub.x}$, $n_{sub.i.sub.y}$, and $n_{sub.i.sub.z}$, step **804** comprises computing the position vector $p = (p_{sub.x}, p_{sub.y}, p_{sub.z})_{sup.T}$ of the cell mesh node **304** in real space **140**, using the following equations:

$$(113) \quad p_x = \sum_{i=1}^4 N_i p_{i_x} + \frac{1}{2} n_x t \quad (\text{Equation 622})$$

$$p_y = \sum_{i=1}^4 N_i p_{i_y} + \frac{1}{2} n_y t \quad (\text{Equation 624})$$

$$p_z = \sum_{i=1}^4 N_i p_{i_z} + \frac{1}{2} n_z t \quad (\text{Equation 626})$$

wherein: ζ is the third of the three coordinates

(e.g., η , ξ , and ζ —FIG. **16**) defining the location of the cell mesh node **304** in isoparametric space

200 in the direction of the main axis **212**; p.sub.i.sub.x, p.sub.i.sub.y, and p.sub.i.sub.z are respectively the x, y, and z components (in the real coordinate system) of the position vectors p.sub.1 . . . p.sub.4; t is the local panel thickness of the sandwich panel **120**.

(114) Referring to FIGS. **21-27**, step **706** of mapping reference unit cell meshes **300** is now described with regard to the method **900** of FIG. **21**, which is directed toward mapping the reference unit cell meshes **300** onto a plurality of hexahedral elements **154** respectively associated with quadratic quadrilateral elements **152** of a driver mesh **142**. FIG. **22** shows an example of a portion of a driver mesh **142** comprised of quadratic quadrilateral elements **152**. As mentioned above, each quadratic quadrilateral element **152** has 4 corner nodes **158** (p.sub.1 . . . p.sub.4) and 4 mid-side nodes **160** (p.sub.5 . . . p.sub.8) in real space **140**. FIG. **23** shows a cube **202** in isoparametric space **200**, and the corresponding location of the nodes p.sub.1 . . . p.sub.8 on the cube **202**. FIG. **24** shows an example of a hexahedral element **154** associated with the quadratic quadrilateral element **152** of FIG. **22**.

(115) Similar to the above-described process of mapping a reference unit cell mesh **300** onto a hexahedral element **154** associated with a linear quadrilateral element **150** through the use of linear basis functions, the method **900** of FIG. **21** describes the process of mapping a reference unit cell mesh **300** onto a hexahedral element **154** associated with a quadratic quadrilateral element **152** through the use of quadratic basis functions. FIG. **25** shows 8 groups of level set contours respectively associated with the 8 quadratic basis functions N.sub.1 . . . N.sub.8 defined on an 8-noded quadratic quadrilateral element **152**. The level set contours indicate the weight of a given corner node **158** across the 8-noded quadratic quadrilateral element **152**. FIG. **27** shows the reference unit cell mesh **300** of FIG. **26** after being mapped onto the hexahedral element **154** (e.g., FIG. **24**), associated with the quadratic quadrilateral **152** (FIG. **22**), belonging to the driver mesh **142**.

(116) Referring to FIGS. **23-27**, the method **900** of mapping the reference unit cell mesh **300** onto a hexahedral element **154** associated with a quadratic quadrilateral element **152** of a driver mesh **142** comprises sequentially performing steps **902**, **904**, and **906** for each quadratic quadrilateral element **152** of a driver mesh **142**. Step **902** comprises computing the nodal averaged normal vectors n.sub.1 . . . n.sub.4 (FIG. **24**) of the corner nodes **158** (p.sub.1 . . . p.sub.4) of the quadratic quadrilateral element **152** in real space **140**, as described above.

(117) After computing the nodal averaged normal vectors n.sub.1 . . . n.sub.4, step **904** comprises computing the mid-side normal vectors (n.sub.5 . . . n.sub.8) of the quadratic quadrilateral element **152** by taking, for each mid-side normal vector, the average of the normal vectors at the corner nodes **158** on opposite ends of the edge **162** upon which the mid-side node **160** lies, using the following equations:

$$(118) \quad n_x = \frac{1}{2}(n_{A_x} + n_{B_x}) \quad (\text{Equation628}) \quad n_y = \frac{1}{2}(n_{A_y} + n_{B_y}) \quad (\text{Equation630})$$

$n_z = \frac{1}{2}(n_{A_z} + n_{B_z})$ (Equation632) wherein: n.sub.A.sub.x, n.sub.A.sub.y and n.sub.A.sub.z are the x, y and z component respectively of the nodal averaged normal vector of the first corner node **158** of the edge **162** on which the mid-side node **160** lies; n.sub.B.sub.x, n.sub.B.sub.y and n.sub.B.sub.z are the x, y and z components respectively of the second corner node **158** of the edge **162** on which the mid-side node **160** lies; and n.sub.x, n.sub.y and n.sub.z are the x, y and z components respectively of the mid-side nodal averaged normal vector.

(119) Step **906** comprises sequentially performing the following computations for each cell mesh node **304** of the reference unit cell mesh **300**. Initially, step **906** includes computing the quadratic basis functions N.sub.1 . . . N.sub.8 of the cell mesh node **304**, using the following equations:

$$(120) \quad N_1 = \frac{1}{4}(1 - \quad)(1 - \quad)(- \quad - 1) \quad (\text{Equation634})$$

$$N_2 = \frac{1}{4}(1 + \quad)(1 - \quad)(- \quad - 1) \quad (\text{Equation636})$$

$$N_3 = \frac{1}{4}(1 + \quad)(1 + \quad)(+ \quad - 1) \quad (\text{Equation638})$$

$$N_4 = \frac{1}{4}(1 - \eta)(1 + \xi)(-1 + \zeta - 1) \quad (\text{Equation 640}) \quad N_5 = \frac{1}{2}(1 - \eta)(1 + \xi)(1 - \zeta) \quad (\text{Equation 642})$$

$$N_6 = \frac{1}{2}(1 + \eta)(1 + \xi)(1 - \zeta) \quad (\text{Equation 644}) \quad N_7 = \frac{1}{2}(1 + \eta)(1 + \xi)(1 - \zeta) \quad (\text{Equation 646})$$

$$N_8 = \frac{1}{2}(1 + \eta)(1 - \xi)(1 - \zeta) \quad (\text{Equation 648})$$

wherein η and ξ (e.g., see FIG. 23) are the coordinates of the cell mesh node **304** in isoparametric space **200**.

(121) After computing the quadratic basis functions $N_{\text{sub.1}} \dots N_{\text{sub.8}}$, step **906** comprises computing the normal vector $\eta = (n_{\text{sub.x}}, n_{\text{sub.y}}, n_{\text{sub.z}})$ of the panel mid-surface **126** at the position of the cell mesh node **304** in real space **140**, using the following equations:

$$n_{\text{sub.x}} = \sum_{\text{sub.i}=1}^{\text{sup.8}} N_{\text{sub.in.sub.i.sub.x}} \quad (\text{Equation 650})$$

$$n_{\text{sub.y}} = \sum_{\text{sub.i}=1}^{\text{sup.8}} N_{\text{sub.in.sub.i.sub.y}} \quad (\text{Equation 652})$$

$$n_{\text{sub.z}} = \sum_{\text{sub.i}=1}^{\text{sup.8}} N_{\text{sub.in.sub.i.sub.z}} \quad (\text{Equation 654})$$

wherein $n_{\text{sub.i.sub.x}}$, $n_{\text{sub.i.sub.y}}$, and $n_{\text{sub.i.sub.z}}$ are respectively the x, y, and z components (in the real coordinate system—e.g., FIG. 24) of the nodal averaged normal vectors $n_{\text{sub.1}} \dots n_{\text{sub.8}}$.

(122) After computing $n_{\text{sub.i.sub.x}}$, $n_{\text{sub.i.sub.y}}$, and $n_{\text{sub.i.sub.z}}$, step **906** comprises computing the position vector $p = (p_{\text{sub.x}}, p_{\text{sub.y}}, p_{\text{sub.z}})_{\text{sup.T}}$ of the cell mesh node **304** in real space **140**, using the following equations:

$$(123) \quad p_x = \sum_{i=1}^8 N_i p_{i_x} + \frac{1}{2} n_x t \quad (\text{Equation 656})$$

$$p_y = \sum_{i=1}^8 N_i p_{i_y} + \frac{1}{2} n_y t \quad (\text{Equation 658})$$

$$p_z = \sum_{i=1}^8 N_i p_{i_z} + \frac{1}{2} n_z t \quad (\text{Equation 660})$$

wherein: $n_{\text{sub.i.sub.x}}$, $n_{\text{sub.i.sub.y}}$, and $n_{\text{sub.i.sub.z}}$ are respectively the x, y, and z components (relative to the real coordinate system) of the nodal averaged normal vectors $n_{\text{sub.1}} \dots n_{\text{sub.8}}$ at the nodes $p_{\text{sub.1}} \dots p_{\text{sub.8}}$ respectively; $p_{\text{sub.i.sub.x}}$, $p_{\text{sub.i.sub.y}}$, and $p_{\text{sub.i.sub.z}}$ are respectively the x, y, and z components (relative to the real coordinate system) of the position vectors $p_{\text{sub.1}} \dots p_{\text{sub.8}}$; t is the local panel thickness of the sandwich panel **120**. Here, thickness t refers to the total thickness of the sandwich panel **120**, including the two face sheets **102**, **104** and the panel infill geometry **166**.

(124) After the mapping step **706** is complete, the method **700** includes step **708** of stitching together the mapped unit cell meshes **302** to form a stitched mesh **130** (e.g., FIGS. 28-29). The stitching process initially includes checking pairs of cell mesh nodes **304** for duplicates that share a common set of coordinates in real space **140**. Duplicate cell mesh nodes **304** will occur along the edges of adjacent (e.g., abutting) mapped unit cell meshes **302**. Upon identifying the duplicates, step **708** further includes deleting one of the duplicates, and updating the cell information to refer to the cell mesh node **304** that is retained.

(125) Referring to FIGS. 28-29, shown is an example of a panel digital representation (e.g., a stitched mesh **130**) of a sandwich panel **120** having a panel geometry comprised of the panel infill geometry **166**, and both of the panel face sheet geometries **170** of the first and second face sheets **102**, **104**, as generated using the above-described method **700**. The panel geometry is comprised of a plurality of mapped unit cell meshes **302**, stitched together into a single mesh of the sandwich panel **120**, and is the result of mapping the reference unit cell meshes **300** onto the hexahedral elements **154** associated with the quadrilateral elements **148** of the driver mesh **142** of FIG. 10. The panel face sheet geometries **170** of the panel digital representation are integrated into the unit cell geometry **210** (e.g., FIG. 26), and are therefore automatically generated upon mapping and stitching.

(126) Referring to FIGS. 30-31, the method **700**, in some examples, includes using the panel digital representation to develop an additive manufacturing program **532** (i.e., a software program—see FIG. 70). The additive manufacturing program **532** can be used by an additive manufacturing apparatus **400** to fabricate the sandwich panel **120**. FIGS. 30-31 show an example of an additive

manufacturing apparatus **400** for fabricating the sandwich panel **120** of FIGS. **28-29**. As mentioned above, additive manufacturing involves the successive application and solidification of layers of material on top of each other.

(127) In the example of FIGS. **30-31**, the additive manufacturing apparatus **400** is configured as a powder bed fusion machine that uses selective laser melting for additive manufacturing. The additive manufacturing apparatus **400** includes a powder supply chamber **412** containing a powder **410**. The powder **410** may be a polymeric powder, a ceramic powder, a metal powder, or any combination thereof. The powder supply chamber **412** is vertically movable via a powder supply piston **416**, which is actuated by a powder supply actuator **418**.

(128) The additive manufacturing apparatus **400** includes a powder leveling apparatus **420** (e.g., a roller) and a build chamber **402**. The build chamber **402** includes a build platform **404** that supports a powder bed **414**. The build platform **404** is vertically movable in a stepwise manner via a build platform piston **406**, which is actuated by a build platform actuator **408**. The additive manufacturing apparatus **400** further includes an energy source **424** or heat source, such as a laser device **426** configured to emit a laser beam **428**. During additive manufacturing of the sandwich panel **120**, the powder leveling apparatus **420** periodically moves across the powder **410** contained within the powder supply chamber **412** and onto the powder bed **414**, to thereby distribute a thin layer of powder **410** onto the powder bed **414**. The excess powder **410** falls into the powder overflow chamber **422**.

(129) The laser beam **428** is moved in a manner to form the cross-sectional shape of a slice of the panel digital representation of the sandwich panel **120**, as shown in FIG. **31**. The laser beam **428** melts and fuses the powder **410** particles into a solid layer in the shape of the slice. The build platform **404** is moved incrementally downwardly, via the build platform piston **406** after which the powder leveling apparatus **420** distributes a new layer of powder **410** on the existing powder **410** in the powder bed **414** of the build chamber **402**. The process is repeated layer by layer until the manufacturing of the sandwich panel **120** is complete. Although the additive manufacturing apparatus **400** is shown and described as a selective laser melting system (e.g., a laser-based powder bed fusion system), any one of a variety of alternative types of additive manufacturing technologies (e.g., electron beam melting, direct energy deposition, supersonic particle deposition, binder jetting, etc.) may be implemented for manufacturing sandwich panels **120** using the presently-disclosed method.

(130) Referring to FIG. **32**, shown is an example of a driver mesh **142**. FIGS. **33-56** show non-exclusive examples of different configurations of the unit cell geometry **210**. Also shown is the panel infill geometry **166** that results from mapping each of the different unit cell geometries **210** onto the driver mesh **142** of FIG. **32**, using the above-described step **706**. As mentioned above, the driver mesh **142** of FIG. **32** is an example of a structured driver mesh **144**, which may be described as a pattern of quadrilateral elements **148** that can be mapped into a square domain with uniform rows and columns.

(131) FIG. **33** shows a unit cell geometry **210** having a unit infill geometry **310** configured as a hexagonal honeycomb **220**, and having a unit face sheet geometry **316** on both the top and bottom ends of the hexagonal honeycomb **220**. FIG. **34** shows a reference unit cell mesh **300** representing the unit cell geometry **210** of FIG. **33**. The reference unit cell mesh **300** includes a unit infill mesh **314**, and a pair of unit face sheet meshes **318** on the top and bottom ends of the unit infill mesh **314**. In the example shown, the reference unit cell mesh **300** is comprised of shell elements **306** (e.g., quadrilateral elements). FIG. **35** shows an example of a surface-based sandwich panel **120** produced by mapping the reference unit cell mesh **300** of FIG. **34** onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** contained in the driver mesh **142** of FIG. **32**, and resulting in a plurality of mapped unit cell meshes **302** collectively forming a panel infill mesh **168** having a panel infill geometry **166**. In FIG. **35**, the panel face sheet meshes **172** representing the first and second face sheets **102**, **104** have been

mitted for clarity.

(132) FIG. 36 shows an example of solid-based sandwich panel 120 produced by mapping a reference unit cell mesh 300 comprised of solid elements 308 (e.g., tetrahedral elements) onto a plurality of hexahedral elements 154 (not shown) respectively associated with the plurality of quadrilateral elements 148 of the driver mesh 142 of FIG. 32. The sandwich panel 120 of FIG. 36 includes the panel infill geometry 166, and the panel face sheet geometry 170 of the first and second face sheets 102, 104.

(133) The sandwich panel 120 of FIG. 36 includes a first face sheet 102, a second face sheet 104, and the panel infill structure 164. The panel infill structure 164 of the sandwich panel 120 comprises an array of unit infill structures 312 arranged in a single layer connecting the first face sheet 102 to the second face sheet 104. As described above, each unit cell structure 206 is configured to fit exactly within a hexahedral element 154 (FIG. 17). In addition, in isoparametric space 200, each unit cell structure 206 can be described as being periodic with respect to at least the η and ξ directions of the isoparametric cube 202 (e.g., FIGS. 16, 23). Furthermore, each unit cell structure 206 has an orthonormal coordinate system having a main axis 212 that, after mapping, is locally normal to the panel mid-surface 126 between the first face sheet 102 and the second face sheet 104. In isoparametric space 200, the main axis 212 is aligned with the ζ direction of an isoparametric cube 202 (FIGS. 16, 23). After mapping and stitching is complete, each unit cell structure 206 is connected to one or more adjacent unit cell structures 206 due to the periodic nature of the hexagonal honeycomb 220.

(134) As shown in FIG. 36, the first face sheet 102 and the second face sheet 104 each have a surface contour that is non-planar. More specifically, the first face sheet 102 and the second face sheet 104 each have non-zero principal curvatures. The first face sheet 102 and the second face sheet 104 do not have the same principal curvatures. Correspondingly, at the infill/face sheet interfaces 110 with the first face sheet 102, the unit infill structures 312 have a larger cross-sectional size or footprint than the footprint of the unit infill structures 312 at the infill/face sheet interfaces 110 with the second face sheet 104. While these footprints are different, the topology of the footprints is the same. Although FIGS. 33-36 show the unit infill geometry 310 having a honeycomb 220 configuration that has flat hexagonal walls, the unit infill geometry 310 may be provided in different configurations. As an example, the hexagonally-arranged walls may be provided with a pattern of holes to facilitate removal of material, such as excess powder following the completion of an additive manufacturing build process.

(135) FIG. 37 shows a unit cell geometry 210 having a unit infill geometry 310 in a gyroid 216 configuration. The unit cell geometry 210 also has a unit face sheet geometry 316 located on each of the top and bottom ends of the gyroid 216. FIG. 38 shows a reference unit cell mesh 300 representing the unit cell geometry 210 of FIG. 37. As mentioned above, the reference unit cell mesh 300 includes a unit infill mesh 314, and a pair of unit face sheet meshes 318 on opposite ends of the unit infill mesh 314. In the example shown, the reference unit cell mesh 300 is comprised of shell elements 306. Alternatively, the reference unit cell mesh 300 can be provided as a solid unit cell comprised of solid elements 308 (e.g., tetrahedral elements), as shown in above-described FIG. 13. FIG. 39 shows an example of a surface-based sandwich panel 120 produced by mapping the reference unit cell mesh 300 of FIG. 38 onto a plurality of hexahedral elements 154 (not shown) respectively associated with the plurality of quadrilateral elements 148 contained in the driver mesh 142 of FIG. 32, and resulting in a plurality of mapped unit cell meshes 302 collectively forming a panel infill mesh 168 having a panel infill geometry 166. In FIG. 39, the panel face sheet meshes 172 representing the first and second face sheets 102, 104 have been omitted for clarity.

(136) FIG. 40 shows an example of a solid-based sandwich panel 120 produced by mapping a reference unit cell mesh 300 comprised of solid elements 308 (FIG. 13) onto a plurality of hexahedral elements 154 (not shown) respectively associated with the plurality of quadrilateral elements 148 of the driver mesh 142 of FIG. 32. The sandwich panel 120 of FIG. 40 includes the

panel infill geometry **166** (i.e., the panel infill structure **164**), and the panel face sheet geometry **170** of the first and second face sheets **102**, **104**. After mapping and stitching is complete, each unit cell structure **206** is connected to one or more adjacent unit cell structures **206** due to the periodic nature of the gyroid **216**.

(137) FIG. **41** shows a unit cell geometry **210** having a unit infill geometry **310** in a Schwarz-P **214** configuration, and having a unit face sheet geometry **316** on both the top and bottom ends of the Schwarz-P **214** configuration. In the example shown, the Schwarz-P **214** configuration has 4 cell branches **224** that are orthogonal to the main axis **212**. Each cell branch **224** terminates at one of the faces of the hexahedral element **154** (e.g., FIG. **17**). FIG. **42** shows a reference unit cell mesh **300** representing the Schwarz-P **214** configuration of the unit cell geometry **210** of FIG. **41**. The reference unit cell mesh **300** includes a unit infill mesh **314**, and a pair of unit face sheet meshes **318** respectively on the top and bottom of the unit infill mesh **314**. In the example shown, the reference unit cell mesh **300** is comprised of shell elements **306**. FIG. **43** shows an example of a surface-based sandwich panel **120** produced by mapping the reference unit cell mesh **300** of FIG. **42** onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** contained in the driver mesh **142** of FIG. **32**, and resulting in a plurality of mapped unit cell meshes **302** collectively forming a panel infill mesh **168** having a panel infill geometry **166**. In FIG. **43**, the panel face sheet meshes **172** representing the first and second face sheets **102**, **104** have been omitted for clarity.

(138) FIG. **44** shows an example of a sandwich panel **120** produced by mapping a solid-based reference unit cell mesh **300** (not shown) onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** of the driver mesh **142** of FIG. **32**. Through the above-described process of stitching to remove duplicate cell mesh nodes **304**, the unit cell structures **206** of FIGS. **43-44** are interconnected to each other at the cell branches **224** due to the periodic nature of the Schwarz-P **214** configuration. Although not shown the Schwarz-P configuration may be provided with holes to facilitate powder removal following the completion of an additive manufacturing build process.

(139) FIG. **45** shows a unit cell geometry **210** having a unit infill geometry **310** in a waffle **222** configuration. The waffle **222** configuration has waffle walls with a circular hole to facilitate powder removal following an additive manufacturing process. FIG. **46** shows a reference unit cell mesh **300** made up of shell elements **306**, and representing the unit cell geometry **210** of FIG. **45**. As mentioned above, the reference unit cell mesh **300** of FIG. **46** includes a unit infill mesh **314**, and a pair of unit face sheet meshes **318** on the top and bottom of the unit infill mesh **314**. FIG. **47** shows an example of a surface-based sandwich panel **120** produced by mapping the reference unit cell mesh **300** of FIG. **46** onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** contained in the driver mesh **142** of FIG. **32**, and resulting in a plurality of mapped unit cell meshes **302** collectively forming a panel infill mesh **168** having a panel infill geometry **166**. In FIG. **47**, the panel face sheet meshes **172** representing the first and second face sheets **102**, **104** have been omitted for clarity.

(140) FIG. **48** shows an example of solid-based sandwich panel **120** produced by mapping a reference unit cell mesh **300** comprised of solid elements **308** (not shown) onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** of the driver mesh **142** of FIG. **32**. The sandwich panel **120** of FIG. **48** includes the panel infill geometry **166**, and the panel face sheet geometry **170** of the first and second face sheets **102**, **104**. After completion of the above-described stitching process, the unit cell structures **206** of FIGS. **47-48** are interconnected due to their periodic nature. Although not shown here, the waffle **222** configuration can be provided without holes in the waffle walls or, alternatively, the waffle **222** configuration can be provided with one or more holes in an arrangement different to what is shown.

(141) FIG. **49** shows a unit cell geometry **210** having a unit infill geometry **310** configured as a body-centered-cubic lattice **218**, and having a unit face sheet geometry **316** on both the top and

bottom ends of the body-centered-cubic lattice **218**. FIG. **50** shows a surface-based reference unit cell mesh **300** representing the unit cell geometry **210** of FIG. **49**. The reference unit cell mesh **300** includes a unit infill mesh **314**, and a pair of unit face sheet meshes **318** on the top and bottom of the unit infill mesh **314**. FIG. **51** shows an example of a surface-based sandwich panel **120** produced by mapping the reference unit cell mesh **300** of FIG. **50** onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** contained in the driver mesh **142** of FIG. **32**, and resulting in a plurality of mapped unit cell meshes **302** collectively forming a panel infill mesh **168** having a panel infill geometry **166**. In FIG. **51**, the panel face sheet meshes **172** representing the first and second face sheets **102**, **104** have been omitted for clarity.

(142) FIG. **52** shows an example of a solid-based sandwich panel **120** produced by mapping a reference unit cell mesh **300** comprised of solid elements **308** (not shown) onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** of the driver mesh **142** of FIG. **32**. The sandwich panel **120** of FIG. **52** includes the panel infill geometry **166**, and the panel face sheet geometry **170** of the first and second face sheets **102**, **104**. After completion of the stitching process to remove duplicate cell mesh nodes, the unit cell structures **206** in FIGS. **51-52** are interconnected due to the periodic nature of the body-centered-cubic lattice **218**. Although shown with holes in FIGS. **49-52**, the body-centered-cubic lattice **218** may be provided without holes or, alternatively, the body-centered-cubic lattice **218** may be provided with one or more holes in an arrangement different to what is shown.

(143) Referring to FIGS. **53-56**, shown in FIG. **53** is an example of the above-described multi-layer unit cell geometry **226**, in which the multi-layer unit infill geometry **230** includes 2 or more side-by-side stacks of unit infill structures **312**. More specifically, the multi-layer unit infill geometry **230** of FIG. **53** is a $3 \times 3 \times 3$ arrangement having 3 layers of unit infill structures **312** stacked on top of each other, and each layer has a 3×3 arrangement of unit infill structures **312**, for a total of 27 unit infill structures **312** included in the multi-layer unit infill geometry **230**. As mentioned above, each unit infill structure **312** is a gyroid **216** configuration, which is triply periodic (i.e., periodic in the η , ξ and ζ directions). The multi-layer unit cell geometry **226** of FIG. **53** has a unit face sheet geometry **316** on both the top and bottom of the multi-layer unit infill geometry **230**. FIG. **54** shows the above-described multi-layer unit cell mesh **228** representing the multi-layer unit cell geometry **226** of FIG. **53**. In the example shown, the reference unit cell mesh **300** is comprised of shell elements **306** (e.g., quadrilateral elements). The multi-layer unit cell mesh **228** includes the multi-layer unit infill mesh **232**, and a pair of unit face sheet meshes **318** respectively on the top and bottom of the multi-layer unit infill mesh **232**.

(144) FIG. **55** shows an example of a surface-based sandwich panel **120** produced by mapping the multi-layer unit cell mesh **228** of FIG. **54** onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** contained in the driver mesh **142** of FIG. **32**. In FIG. **55**, the first and second face sheets **102**, **104** have been omitted for clarity. FIG. **56** shows an example of a solid-based sandwich panel **120** produced by mapping a multi-layer unit cell mesh **228** comprised of solid elements (not shown) onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** of the driver mesh **142** of FIG. **32**. The sandwich panel **120** of FIG. **56** includes the panel infill geometry **166**, and the panel face sheet geometry **170** of the first and second face sheets **102**, **104**. After completion of the stitching process to remove duplicate cell mesh nodes **304**, the multi-layer unit cell meshes **228** in FIGS. **55-56** are interconnected due to their periodic nature.

(145) Referring to FIGS. **57-59**, shown in FIG. **57** is an example of a driver mesh **142** configured as an unstructured driver mesh **146**. As mentioned above, the quadrilateral elements **148** of an unstructured driver mesh **146** are arranged in a pattern which cannot be mapped back into a square domain with rows and columns. In the example of FIG. **57**, an unstructured driver mesh **146** is a convenient way of constructing a mesh on a panel mid-surface **126** containing a round hole. FIG.

58 shows an example of a Schwarz-P **214** configuration of a unit cell geometry **210**. FIG. **59** shows a panel infill geometry **166** generated by mapping the unit cell geometry **210** of FIG. **58**, as represented by the reference unit cell mesh **300** shown in FIG. **42**, onto the unstructured driver mesh **146** of FIG. **57**. For an unstructured driver mesh **146**, the reference unit cell mesh **300** is limited to a unit cell geometry **210** in which the pattern of cell mesh nodes **304** on the 4 square faces **204** respectively at $\eta=1$, $\eta=-1$, $\xi=1$ and $\xi=-1$ are identical. Examples of this include a Schwarz-P **214** configuration (e.g., FIG. **38**), the body-centered-cubic lattice **218** configuration (e.g., FIG. **50**), and a waffle **222** configuration (e.g., FIG. **46**).

(146) Referring to FIGS. **60-62**, shown in FIG. **60** is an example of a driver mesh **142** that is spatially graded. In a spatially-graded driver mesh **142**, the density of the driver mesh nodes **156** is spatially graded over certain regions of the driver mesh **142**. Increasing the node density locally, will locally reduce the size of the quadrilateral elements **148**, while decreasing the node density locally, will locally increase the size of the quadrilateral elements **148**. Increasing the node density may be desirable in regions that are subjected to high loads, and will result in a panel infill geometry **166** having a higher density of unit cell structures **206** in such regions. However, the node density may be locally adjusted (e.g., increased or decreased) based on any one of a variety of physical quantities (e.g., heat, electrical current, etc.) to which the sandwich panel **120** may be subjected.

(147) In FIG. **60**, the driver mesh **142** represents the panel mid-surface **126** of a sandwich panel **120** (FIG. **62**) having a cylindrical portion **450** protruding upwardly from a circular flange **452** (i.e., base). However, in the example shown, only half of the driver mesh **142** and sandwich panel **120** are shown for clarity. The cylindrical portion **450** and the flange **452** are connected by a radius portion **454**. The density of the driver mesh nodes **156** is higher in the radius portion **454** than in the cylindrical portion **450** or the flange **452**. In FIG. **60**, the node density gradually decreases in the direction of the arrows **157**. For example, the node density gradually decreases in the direction toward the free end of the cylindrical portion **450**. Likewise, the node density gradually decreases in the direction toward the outer perimeter of the circular flange **452**. In addition, the node density on the right-hand side of the cylindrical portion **450** is higher than the node density on the left-hand side of the cylindrical portion **450**.

(148) FIG. **61** shows a unit cell geometry **210** having a unit infill geometry **310** configured as a waffle **222** with center holes. FIG. **62** is a cut-away view of a sandwich panel **120** having a panel infill geometry **166** generated by mapping the unit cell geometry **210** of FIG. **61** onto a plurality of hexahedral elements **154** (not shown) respectively associated with the plurality of quadrilateral elements **148** contained in the spatially-graded driver mesh **142** of FIG. **60**. The increased node density in the radius portion **454** of the driver mesh **142** results in an increased density of unit cell structures **206** at that location in the sandwich panel **120**. The locally increased density of unit cell structures **206** allows the radius portion **454** to handle higher loads at that location. Conversely, the decreasing node density of the driver mesh **142** (FIG. **60**—along the direction of the arrows **157**) in the cylindrical portion **450** and in the flange **452** results in a gradually decreasing density of unit cell structures **206** along the direction of the arrows **157** in FIG. **62**. In addition, the higher node density on the right-hand side of the cylindrical portion **450** in FIG. **62** results in a higher density of unit cell structures **206** at that location than on the left-hand side of the cylindrical portion **450** in FIG. **62**. That is to say, the density of unit cell structures **206** decreases in a circumferentially-counterclockwise direction, along the direction of the arrow **157**. As shown in FIG. **62**, despite the spatial grading, the sandwich panel **120** has a consistent topology at the infill/face sheet interfaces **110** between the panel infill structure **164** and the first and second face sheets **102**, **104**.

(149) Referring to FIGS. **63-66**, shown in FIG. **63** is an example of a constant-thickness sandwich panel **122**. The constant-thickness sandwich panel **122** has a panel infill structure **164** comprised of unit cell structures **206** in which the unit infill geometry **310** has a Schwarz-P **214** configuration. The panel thickness t is defined by the spacing between the outermost surfaces of the first face

sheet **102** and the second face sheet **104**. In FIG. **63**, the panel thickness t is uniform throughout all regions of the constant-thickness sandwich panel **122**.

(150) FIG. **64** shows an example of a variable-thickness sandwich panel **124**, also having a panel infill structure **164** comprised of unit cell structures **206** in which the unit infill geometry **310** has a Schwarz-P **214** configuration. The first face sheet **102** and the second face sheet **104** are spaced apart from each other at a non-constant spacing. More specifically, the panel thickness t gradually tapers down from the center of the variable-thickness sandwich panel **124** toward each of the opposing ends of the variable-thickness sandwich panel **124**. Although the panel infill geometry **166** of the sandwich panels in FIGS. **63-64** have the same number of unit cell structures **206**, the unit cell structures **206** in FIG. **64** are of varying heights, in correspondence with the varying panel thickness t .

(151) Referring to FIG. **65-66**, shown in FIG. **65** is another example of a constant-thickness sandwich panel **122**. FIG. **66** shows a variable-thickness sandwich panel **124** in which the local panel thickness t is uniform along the right-hand side of the variable-thickness sandwich panel **124**, and is uniform but reduced along the left-hand side of the variable-thickness sandwich panel **124**, with a smooth transition of thickness between these two regions. In both examples, the panel infill structure **164** is comprised of unit cell structures **206** in which the unit infill geometry **310** has a body-centered-cubic lattice **218** configuration.

(152) Referring to FIGS. **67-69**, shown in FIG. **67** is an example of a nose section of a fuselage **462** of an aircraft **460**. The rectangular area identified by reference numeral **68** is a region of the fuselage **462** containing an existing panel **464** to be removed and replaced with a replacement sandwich panel **468** to be manufactured using the presently-disclosed method **700**. In order to manufacture the replacement sandwich panel **468**, it is necessary to provide a driver mesh **142** that represents the size, shape, and contour of the existing panel **464**.

(153) Referring briefly to FIG. **8**, in the above-mentioned example, step **702** of providing the driver mesh **142** comprises acquiring a panel surface **466** of the existing panel **464**. In one example, the panel surface **466** can be acquired by scanning the panel surface **466** using a laser scanner (not shown) or other suitable scanning device. After generating a point cloud (not shown) representing the panel surface **466**, step **702** includes converting the point cloud into a surface digital representation (not shown) of the panel surface **466**. The surface digital representation may be edited to remove extraneous features that are non-representative of the panel surface **466**. After generating the surface digital representation, step **702** includes constructing a mid-surface digital representation (not shown) of the panel mid-surface **126** of the existing panel **464**, based on the surface digital representation of the panel surface **466**. The panel mid-surface **126** of the existing panel **464** may be determined by offsetting the surface digital representation by an amount equal to half the distance of the local panel thickness t (which may vary across the sandwich panel **120**) of the existing panel **464**. Step **702** then includes meshing the mid-surface digital representation of the panel mid-surface **126**, to result in a driver mesh **142**, as shown in FIG. **68**. As described above, the driver mesh **142** is comprised of quadrilateral elements **148** arranged in the desired pattern.

(154) Referring to FIG. **69**, shown is an example of a replacement sandwich panel **468**, having a panel infill geometry **166** generated by mapping a unit cell geometry **210** (not shown) onto the driver mesh **142** of FIG. **68**. The replacement sandwich panel **468** includes first and second face sheets **102**, **104**, interconnected by the panel infill geometry **166**. In the example shown, the panel infill geometry **166** is comprised of uniformly spaced unit cell structures **206**, each having a gyroid **216** configuration. However, the panel infill geometry **166** of a replacement sandwich panel **468** can be generated by mapping any one of a variety of other configurations (e.g., honeycomb, waffle, Schwarz-P, body-centered-cubic, etc.) of unit cell geometries **210** onto the driver mesh **142**. Additionally, the panel infill geometry **166** of a replacement sandwich panel **468** can be spatially graded as necessary for strength purposes or for other reasons.

(155) Referring now to FIG. **70**, the above-described methods **700**, **800**, and **900**, in whole or in

part, may be performed in a computer-implemented process, such as on a processor-based system **500** or other suitable computer system. The processor-based system **500** is configured to perform computer readable program instructions **530**. The computer readable program instructions **530** are provided to, or are loaded onto, the processor-based system **500** in order to implement the above-described operations or steps of methods **700**, **800**, and **900**. In a non-limiting example, the processor-based system **500** and/or the computer readable program instructions **530** generate a panel infill geometry **166** for a sandwich panel **120**.

(156) The block diagram of FIG. **70** illustrates the processor-based system **500** in an advantageous example for generating a panel infill geometry **166** for a sandwich panel **120**. The processor-based system **500** includes a data communication path **502** (e.g., a data link) to communicatively couple one or more components, to facilitate transfer of data between such components. The communication path **502** comprises one or more data buses, or any other suitable communication path **502** that facilitates the transfer of data between the components and devices of the processor-based system **500**. In a non-limiting example, the components include one or more of a processor **504**, a memory device **506**, a non-volatile storage device **508**, a communications device **512**, an input/output device **510**, a display device **514**, a unit cell mapping module **516**, a panel export module **518**, and an additive manufacturing program generator **520**.

(157) The memory device **506** is configured to store one or more driver meshes **142**, each representing a panel mid-surface **126** of a sandwich panel **120**. As described above, each driver mesh **142** is comprised of a plurality of quadrilateral elements **148**. The quadrilateral elements **148** are linear quadrilateral elements **150**, or quadratic quadrilateral elements **152**, as described above. Each driver mesh **142** is stored either as a structured driver mesh **144**, or as an unstructured driver mesh **146**, as is also described above. In some examples, the driver mesh **142** is spatially graded over at least one region of the panel mid-surface **126**, to either refine or coarsen the driver mesh **142** in such regions.

(158) In addition to storing one or more driver meshes **142**, the memory device **506** is configured to store one or more reference unit cell meshes **300**, each having a different unit cell geometry **210**, and each configured to fit exactly within a cube **202**, as described above. Examples of configurations of the reference unit cell meshes **300** include, but are not limited to, Schwarz-P **214** (with or without holes), gyroid **216**, body-centered-cubic lattice **218** (with or without holes), honeycomb **220** (with or without holes), or waffle **222** (with or without holes). The configurations of reference unit cell mesh **300** are periodic with respect to at least the η and ξ directions of the isoparametric cube **202**. The periodic nature of a reference unit cell mesh **300**, together with the above-described stitching operation, enables adjacent mapped unit cell meshes **302** to be connected to each other, to thereby result in a continuous sandwich panel mesh **132**. Some configurations of the reference unit cell mesh **300** can include one or more cell branches **224**, each of which terminates at a square face **204** of the cube **202**. When mapped onto the hexahedral elements **154** (e.g., FIG. **17**) associated with the quadrilateral elements **148** of a driver mesh **142**, and after being stitched together, the cell branches **224** of adjacent mapped unit cell meshes **302** are interconnected to form a continuous sandwich panel mesh **132** (i.e., a stitched mesh **130**).

(159) Referring still to FIG. **70**, the processor-based system **500** includes a unit cell mapping module **516** configured to map a plurality of the reference unit cell meshes **300** respectively onto a plurality of hexahedral elements **154** respectively associated with the plurality of quadrilateral elements **148**. The unit cell mapping module **516** performs the above-described mapping operation by applying basis functions to the driver mesh nodes **156** of the plurality of quadrilateral elements **148** in a manner causing adjustment of the size and shape of the plurality of reference unit cell meshes **300** to fit respectively within the plurality of hexahedral elements **154**, and resulting in a plurality of mapped unit cell meshes **302** collectively forming a panel infill geometry **166** interconnecting a first face sheet **102** to a second face sheet **104**. The unit cell mapping module **516** maps the reference unit cell meshes **300** using either linear basis functions or quadratic basis

functions, depending on whether the driver mesh **142** is made up of linear quadrilateral elements **150** or quadratic quadrilateral elements **152**, as described above.

(160) FIG. **71** shows a non-limiting example of a user interface **534** of the processor-based system **500**. In the example shown, the user interface **534** provides the ability to load a “Driver Mesh File,” which defines a driver mesh **142** of quadrilateral elements **148**. In addition, the user interface **534** provides the ability to select whether a driver mesh **142** is provided with linear quadrilateral elements **150** or quadratic quadrilateral elements **152**. In addition to loading a driver mesh **142**, the user interface **534** provides the ability to select a “Cell Geometry” from among a plurality of different configurations of unit cell geometries **210** including, but not limited to, the above-described Schwarz-P, gyroid, body-centered-cubic lattice, honeycomb, waffle configurations, and may also include multi-layer unit cell geometries.

(161) Referring still to FIG. **71**, the user interface **534** provides the ability to specify the panel thickness t , and optionally to generate some combination of the panel infill geometry **166**, the top face sheet geometry **170**, and the bottom face sheet geometry **170**, in lieu of generating the full sandwich panel geometry. The user interface **534** also provides a means for controlling the alignment of the orientation of a unit cell geometry **210** with respect to a given quadrilateral element **148**. For example, in at least one configuration, the user can specify alignment of the η axis or ξ axis of the reference unit cell mesh **300** with the x , y or z axis of the real space **140**. In at least one example of the user interface **534**, the user can specify alignment of the η axis or ξ axis with the vectoral average of the first and third edges **162**, or the vectoral average of the second and forth edges **162** of the quadrilateral element **148** of the driver mesh **142** (e.g., FIG. **10**). The above-noted selections are processed by the unit cell mapping module **516**.

(162) After making the above-described selections, a user presses a button (or functional equivalent) on the user interface **534**, causing the processor-based system **500** to generate a sandwich panel mesh **132** (i.e., a stitched mesh **130**) having a panel infill geometry **166** interconnecting first and second face sheets **102**, **104**. The processor-based system **500** is configured to convert the sandwich panel mesh **132** into a panel digital representation of the sandwich panel **120**. In some examples, the processor-based system includes an additive manufacturing program generator **520** (FIG. **70**). The additive manufacturing program generator **520** is configured to use the panel digital representation of the sandwich panel **120** to generate an additive manufacturing program **532** (i.e., a software program). The additive manufacturing program **532** is outputted to an additive manufacturing apparatus **400** (e.g., FIGS. **30-31**) configured to manufacture a physical version of the sandwich panel **120**, as described above.

(163) Referring still to FIG. **70**, in some examples, the processor-based system **500** includes a panel export module **518** configured to generate a digital representation of the sandwich panel mesh **132** (i.e., a stitched mesh **130**) as an export file that is suitable as an input file that can be read by a numerical analysis program. Such a numerical analysis program is configured to perform a simulation on the sandwich panel mesh **132**. For example, the simulation can determine the distribution of a physical quantity (e.g., displacement, strain, pressure, temperature, etc.) across the sandwich panel mesh **132** in response to the application of a given combination of loads (e.g., mechanical load, thermal load, etc.), constraints (e.g., regions of the sandwich panel **120** that are restrained in some manner, and/or subjected to contact with other bodies), and material properties (e.g., mechanical properties, thermal properties, etc.). In at least one example, the numerical analysis performed on the sandwich panel mesh **132** is a structural simulation employing the finite element method, such as simulating the stress and deflection of the sandwich panel **120** when loaded during a 3-point bending test.

(164) The display device **514** is optionally employed to graphically display any one of the driver meshes **142** and/or reference unit cell meshes **300** stored in the memory device **506**. In addition, the display device **514** is configured to graphically display the sandwich panel mesh **132** resulting from mapping a reference unit cell mesh **300** onto a plurality of hexahedral elements **154** respectively

associated with the plurality of quadrilateral elements **148** contained in the driver mesh **142**. The display device **514** receives the above-noted data from one or more of the corresponding components of the processor-based system **500** via the communication path **502**.

(165) The processor-based system **500** includes the processor **504** for executing instructions of computer readable program instructions **530** installed into the memory device **506**. Alternatively, the processor-based system **500** comprises a multi-processor core having two or more integrated processor cores. Even further, the processor **504** may comprise a main processor and one or more secondary processors integrated on a chip. The processor **504** may also comprise a multi-processor system having a plurality of similarly configured processors.

(166) Referring still to FIG. **70**, the processor-based system **500** includes the memory device **506**, which may comprise one or more volatile or non-volatile storage devices **508**. However, the memory device **506** can comprise any hardware device for storing data. For example, the memory device **506** can comprise a random access memory or a cache of an interface and/or integrated memory controller hub included in the communication path **502**. The memory device **506** can permanently and/or temporarily store any one of a variety of different types of data, computer readable code or program instructions, or any other type of information. The storage device **508** can be provided in a variety of configurations including, but not limited to, a flash memory device, a hard drive, an optical disk, a hard disk, a magnetic tape or any other suitable example for long-term storage. In addition, the storage device **508** can comprise a removable device such as a removable hard drive.

(167) The processor-based system **500** can additionally include one or more of the input/output devices **510** to facilitate the transfer of data between components connected to the processor-based system **500**. An input/output device **510** can be directly and/or indirectly coupled to the processor-based system **500**. The input/output device **510** facilitates user-input by means of a peripheral device such as a keyboard, a mouse, a joystick, a touch screen, and any other suitable device for inputting data to the processor-based system **500**. The input/output device **510** can further include an output device for transferring data representative of the output of the processor-based system **500**. For example the input/output device **510** can comprise a display device **514** such as a computer monitor or computer screen for displaying results of data processed by the processor-based system **500**. The input/output device **510** can optionally include a printer or fax machine for printing a hardcopy of information processed by the processor-based system **500**.

(168) Referring still to FIG. **70**, the processor-based system **500** can include one or more communications devices **512** to facilitate communication of the processor-based system **500** within a computer network and/or with other processor-based systems. Communication of the processor-based system **500** with a computer network or with other processor-based systems can be by wireless means and/or by hardwire connection. For example, the communications device **512** can comprise a network interface controller to enable wireless or cable communication between the processor-based system **500** and a computer network. The communications device **512** can also comprise a modem and/or a network adapter, or any one of a variety of alternative devices for transmitting and receiving data.

(169) One or more of the operations of the above-described methods **700**, **800**, **900** are performed by the processor **504** and/or by one or more of the unit cell mapping module **516**, the panel export module **518**, and/or the additive manufacturing program generator **520**, using the computer readable program instructions **530**. The computer readable program instructions **530** can comprise program code, which can include computer usable program code and computer readable program code. The computer readable program instructions **530** are read and executed by the processor **504**. The computer readable program instructions **530** enable the processor **504** to perform one or more operations of the above-described examples associated with generating the panel infill geometry **166** of a sandwich panel **120**.

(170) Referring still to FIG. **70**, the computer readable program instructions **530** include operating

instructions for the processor-based system **500**, and further include applications and programs. The computer readable program instructions **530** can be contained within and/or loaded onto one or more of a memory device **506** and/or a storage device **508** for execution by the processor **504** and/or by the unit cell mapping module **516**, the panel export module **518**, and/or the additive manufacturing program generator **520**. As indicated above, the memory device **506** and/or storage device **508** can be communicatively coupled to one or more of the remaining components illustrated in FIG. **70** through the communication path **502**.

(171) The computer readable program instructions **530** can be contained on tangible or non-tangible, transitory or non-transitory computer readable media **524** and which can be loaded onto or transferred to the processor-based system **500** for execution by the processor **504**. The computer readable program instructions **530** and the computer readable media **524** comprise a computer program product **522**. In an example, the computer readable media **524** can comprise computer readable storage media **526** and/or computer readable signal media **528**.

(172) The computer readable storage media **526** can comprise a variety of different examples including, but not limited to, optical disks and magnetic disks that may be loaded into a drive, a flash memory device, or other storage device or hardware for transfer of data onto a storage device such as a hard drive. The computer readable storage media **526** can be non-removably installed on the processor-based system **500**. The computer readable storage media **526** can comprise any suitable storage media including, without limitation, a semiconductor system or a propagation medium. In this regard, the computer readable storage media **526** can comprise electronic media, magnetic media, optical media, electromagnetic media, and infrared media. For example, the computer readable storage media **526** can comprise magnetic tape, a computer diskette, random access memory, and read-only memory. Non-limiting examples of optical disks include compact disks—read only memory, compact disks—read/write, and digital video disks (i.e., digital versatile disks).

(173) The computer readable signal media **528** can contain the computer readable program instructions **530**, and can be provided in a variety of data signal configurations including, but not limited to, an electromagnetic signal and an optical signal. Such data signals can be transmitted by any suitable communications link including by wireless or hardwire means. For example, the hardwire means may comprise an optical fiber cable, a coaxial cable, a signal wire, and any other suitable means for transmitting the data by wireless or by physical means.

(174) Referring still to FIG. **70**, the computer readable signal media **528** can facilitate the downloading of the computer readable program instructions **530** to the non-volatile storage or other suitable storage or memory device for use within processor-based system **500**. For example, the computer readable program instructions **530** contained within the computer readable storage media **526** can be downloaded to the processor-based system **500** over a computer network from a server or a client computer of another system.

(175) Any one of a variety of different examples of the processor-based system **500** can be implemented using any hardware device or system capable of executing the computer readable program instructions **530**. For example, the processor **504** can comprise a hardware unit configured for performing one or more particular functions wherein the computer readable program instructions **530** for performing the functions are pre-loaded into the memory device **506**.

(176) In an example, the processor **504** can comprise an application specific integrated circuit, a programmable logic device, or any other hardware device configured to perform one or more specific functions or operations. For example, a programmable logic device may be temporarily or permanently programmed to perform one or more of the operations related to generating the panel infill geometry **166** of a sandwich panel **120**. The programmable logic device can comprise a programmable logic array, programmable array logic, a field programmable logic array, and a field programmable gate array and any other suitable logic device, without limitation. In an example, the computer readable program instructions **530** can be operated by the processor **504** and/or by other

devices, including one or more hardware units in communication with the processor **504**. Certain portions of the computer readable program instructions **530** can be run by the one or more hardware units.

(177) Many modifications and other versions and examples of the disclosure will come to mind to one skilled in the art to which this disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. The versions and examples described herein are meant to be illustrative and are not intended to be limiting or exhaustive. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, are possible from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims. The present disclosure is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A method of generating a panel infill geometry of a sandwich panel, comprising: accessing, through a user interface of a computer, a driver mesh file stored in a memory device of the computer, the driver mesh file comprising a three-dimensional driver mesh representing a panel mid-surface of a sandwich panel that is non-planar, and the three-dimensional driver mesh is comprised of a plurality of quadrilateral elements; accessing, through the user interface, a unit cell geometry file stored in the memory device, the unit cell geometry file comprising a three-dimensional reference unit cell mesh configured to fit exactly within a three-dimensional cube, and the three-dimensional reference unit cell mesh is comprised of a three-dimensional unit infill mesh interconnecting a pair of unit face sheet meshes at opposite ends of the three-dimensional unit infill mesh, and the three-dimensional unit infill mesh and the unit face sheet meshes are made up of a plurality of cell mesh elements and associated cell mesh nodes; and causing a processor of the computer to perform the following, in response to a command received through the user interface: mapping the cell mesh nodes of a plurality of the three-dimensional reference unit cell meshes respectively onto a plurality of three-dimensional hexahedral elements respectively associated with the plurality of quadrilateral elements, through the use of basis functions defined on each of the plurality of quadrilateral elements in a manner causing adjustment of a size and shape of the plurality of the three-dimensional reference unit cell meshes to conform respectively to the plurality of three-dimensional hexahedral elements in a manner forming a plurality of mapped three-dimensional unit cell meshes collectively forming a panel infill mesh having a panel infill geometry interconnecting a pair of panel face sheet meshes and which is non-planar in a static state and having a consistent topology at each interface location on the infill geometry where the plurality of mapped three-dimensional unit cell meshes are connected to the pair of panel face sheet meshes, and all the interface locations where the mapped three-dimensional unit cell meshes transition into the panel face sheet meshes have a footprint of the same shape, regardless of the physical location of the interface location on the sandwich panel; stitching together the plurality of mapped three-dimensional unit cell meshes to form a stitched mesh, by checking pairs of cell mesh nodes for duplicates, deleting one of the duplicates, and updating cell information to refer to the cell mesh node that is retained; and constructing a panel digital representation of the sandwich panel based on the stitched mesh and which is non-planar in the static state and comprising the pair of panel face sheets interconnected by the panel infill geometry having the consistent topology.
2. The method of claim 1, wherein the three-dimensional reference unit cell mesh is a multi-layer unit cell mesh comprised of: two or more layers of three-dimensional unit infill meshes stacked on top of each other; and a pair of unit face sheet meshes respectively located on opposite ends of the multi-layer three-dimensional unit cell mesh.

3. The method of claim 1, further comprising generating the three-dimensional reference unit cell mesh by performing one of the following: constructing the three-dimensional reference unit cell mesh as one or more trimmed parametric surfaces, followed by meshing the one or more trimmed parametric surfaces, to thereby generate a three-dimensional reference unit cell mesh comprised of shell elements; constructing the three-dimensional reference unit cell mesh as a solid structure represented by one or more trimmed parametric surfaces enclosing a volume, followed by meshing the volume, to thereby generate a three-dimensional reference unit cell mesh comprised of solid elements.
4. The method of claim 1, further comprising: constructing the three-dimensional reference unit cell mesh to contain infill geometry that is periodic with respect to two or more principal directions of the three-dimensional cube.
5. The method of claim 1, wherein mapping a plurality of three-dimensional reference unit cell meshes onto the three-dimensional driver mesh comprises one of the following: using linear basis functions to map the plurality of three-dimensional reference unit cell meshes onto a plurality of three-dimensional hexahedral elements respectively associated with the plurality of quadrilateral elements of the three-dimensional driver mesh; using quadratic basis functions to map the plurality of three-dimensional reference unit cell meshes onto a plurality of three-dimensional hexahedral elements respectively associated with the plurality of quadrilateral elements of the three-dimensional driver mesh.
6. The method of claim 1, further comprising generating the three-dimensional driver mesh and generating the three-dimensional reference unit cell mesh by performing one of the following two scenarios: generating the three-dimensional driver mesh as a structured three-dimensional driver mesh, and generating the three-dimensional reference unit cell mesh to have a unit cell geometry of one of the following configurations: Schwarz-P, gyroid, body-centered-cubic lattice, honeycomb, or waffle; generating the three-dimensional driver mesh as an unstructured three-dimensional driver mesh, and generating the three-dimensional reference unit cell mesh to have a unit cell geometry of one of the following configurations: Schwarz-P, body-centered-cubic lattice, or waffle.
7. The method of claim 1, wherein: the three-dimensional driver mesh has a density that is spatially graded over at least one region of the mid-surface.
8. The method of claim 1, further comprising generating the three-dimensional driver mesh by performing one of the following: acquiring a panel surface of an existing panel, to result in a point cloud representing the panel surface; converting the point cloud into a digital representation of the panel surface; constructing a digital representation of the mid-surface of the existing panel, based on the digital representation of the panel surface; and meshing the digital representation of the mid-surface, thereby obtaining the three-dimensional driver mesh.
9. A processor-based system configured to perform computer readable program instructions for generating a panel geometry of a sandwich panel, comprising: a user interface; a memory device containing the computer readable program instructions and configured to store: a driver mesh file, accessible through the user interface, and comprising a three-dimensional driver mesh representing a panel mid-surface of a sandwich panel that is non-planar, the three-dimensional driver mesh is comprised of a plurality of quadrilateral elements; a unit cell geometry file, accessible through the user interface, and comprising a three-dimensional reference unit cell mesh configured to fit exactly within a three-dimensional cube, the three-dimensional reference unit cell mesh is comprised of a three-dimensional unit infill mesh interconnecting a pair of unit face sheet meshes, and the three-dimensional unit infill mesh and the unit face sheet meshes are made up of a plurality of cell mesh elements and associated cell mesh nodes; and a processor configured to execute the computer readable program instructions in response to a command received through the user interface, and execution of the instructions causes the processor to perform functions comprising: mapping the cell mesh nodes of a plurality of the three-dimensional reference unit cell meshes respectively onto a plurality of three-dimensional hexahedral elements respectively associated with

the plurality of quadrilateral elements, through the use of basis functions defined on each of the plurality of quadrilateral elements in a manner causing adjustment of a size and shape of the plurality of the three-dimensional reference unit cell meshes to conform respectively to the plurality of three-dimensional hexahedral elements in a manner forming a plurality of mapped three-dimensional unit cell meshes collectively forming a panel infill mesh having a panel infill geometry interconnecting a pair of panel face sheet meshes and which is non-planar in a static state and having a consistent topology at each interface location on the infill geometry where the plurality of mapped three-dimensional unit cell meshes are connected to the pair of panel face sheet meshes, and all the interface locations where the mapped three-dimensional unit cell meshes transition into the panel face sheet meshes have a footprint of the same shape, regardless of the physical location of the interface location on the sandwich panel; stitching together the plurality of mapped three-dimensional unit cell meshes to form a stitched mesh, by checking pairs of cell mesh nodes for duplicates, deleting one of the duplicates, and updating cell information to refer to the cell mesh node that is retained; constructing a panel digital representation of the sandwich panel based on the stitched mesh is non-planar in the static state.

10. The processor-based system of claim 9, wherein: the three-dimensional reference unit cell mesh is periodic with respect to two or more principal directions of the three-dimensional cube.

11. The processor-based system of claim 9, wherein: the processor is configured to map the plurality of the three-dimensional reference unit cell meshes onto a plurality of three-dimensional hexahedral elements respectively associated with the plurality of quadrilateral elements using one of linear basis functions or quadratic basis functions.

12. The processor-based system of claim 9, wherein the three-dimensional driver mesh and the three-dimensional reference unit cell mesh are stored according to one of the following two scenarios: the three-dimensional driver mesh is stored as a structured three-dimensional driver mesh, and the three-dimensional reference unit cell mesh is stored as a unit cell geometry having one of the following configurations: Schwarz-P, gyroid, body-centered-cubic lattice, honeycomb, or waffle; the three-dimensional driver mesh is stored as an unstructured three-dimensional driver mesh, and the three-dimensional reference unit cell mesh is stored as a unit cell geometry having one of the following configurations: Schwarz-P, body-centered-cubic lattice, or waffle.

13. The processor-based system of claim 9, further comprising: a panel export module configured to export an input file which can be read by a numerical analysis program configured to perform a simulation on a sandwich panel mesh represented by a pair of face sheets and the panel infill geometry, each defined by a plurality of shell elements or solid elements.

14. The processor-based system of claim 9, wherein the three-dimensional reference unit cell mesh is a multi-layer unit cell mesh comprised of: two or more layers of three-dimensional unit infill meshes stacked on top of each other; and a pair of unit face sheet meshes respectively located on opposite ends of the multi-layer unit cell mesh.

15. The processor-based system of claim 9, wherein: the processor is configured to generate the three-dimensional reference unit cell mesh by constructing the three-dimensional reference unit cell mesh as one or more trimmed parametric surfaces, followed by meshing the one or more trimmed parametric surfaces, to thereby result in the three-dimensional reference unit cell mesh comprised of shell elements.

16. The processor-based system of claim 9, wherein: the processor is configured to generate the three-dimensional reference unit cell mesh by constructing the three-dimensional reference unit cell mesh as a solid structure represented by one or more trimmed parametric surfaces enclosing a volume, followed by meshing the volume, to thereby result in the three-dimensional reference unit cell mesh composed of solid elements.

17. The processor-based system of claim 9, wherein: the three-dimensional driver mesh has a density that is spatially graded over at least one region of the mid-surface.

18. The processor-based system of claim 9, wherein the processor is configured to: receive a point

cloud representing a panel surface of an existing panel; convert the point cloud into a digital representation of the panel surface; construct a digital representation of the mid-surface of the existing panel, based on the digital representation of the panel surface; and mesh the digital representation of the mid-surface, thereby obtaining the three-dimensional driver mesh.

19. The processor-based system of claim 9, wherein: the three-dimensional reference unit cell mesh has cell branches, each of which terminates at a square face of the cube.

20. The processor-based system of claim 9, where the user interface is configured to allow for specifying a local panel thickness of the sandwich, defined as a total thickness of the sandwich panel including the pair of face sheets and the panel infill geometry.

21. The processor-based system of claim 9, wherein: the user interface is configured to allow accessing and selecting the driver mesh file from a plurality of driver mesh files stored in the memory.

22. The processor-based system of claim 9, wherein: the user interface is configured to allow accessing and selecting the unit cell geometry file from a plurality unit cell geometry files stored in the memory.
